

Unsupervised Learning

Machine Learning Project

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Project Overview

The Unsupervised Learning Project is a Machine Learning project that works with data without predefined labels. The algorithm finds patterns, groups, and relationships in the given data automatically. The main goal is to understand how unsupervised learning can be used to discover useful information from a dataset.

Objectives

1 Understand the concept of Unsupervised Learning

2 Analyze data without predefined labels

3 Find similar groups of data

4 Identify patterns in the dataset

5 Apply clustering techniques

6 Visualize the results

What is Unsupervised Learning?

Unsupervised Learning is a type of Machine Learning where the model learns from unlabeled data. Unlike supervised learning, there is no predefined output or target value. The algorithm tries to find hidden patterns and groups in the data.



Algorithm Used: K-Means Clustering

K-Means Clustering is used to divide the dataset into different groups called clusters. Each cluster contains data points that are similar to each other.

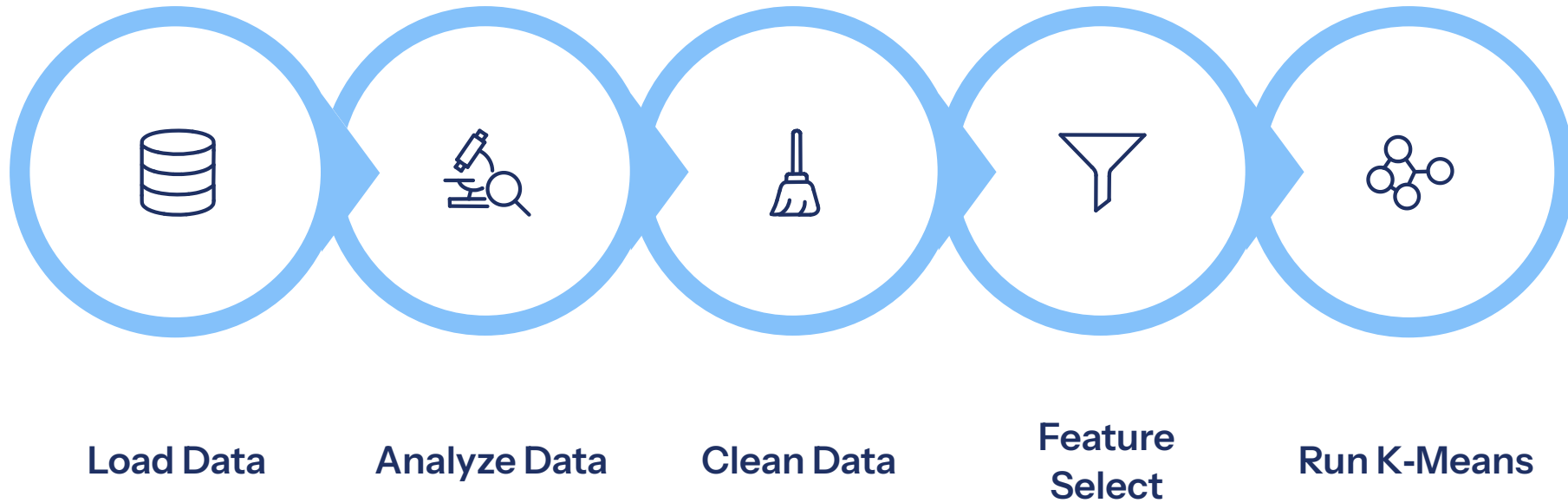




Example: Customer Segmentation

If we have customer data, K-Means can group customers based on their age, income, spending behaviour, and purchase history. This helps businesses understand different types of customers.

How the Project Works



1. Load the dataset 2. Understand and analyse the data 3. Clean the data if required 4. Select important features 5. Apply the K-Means clustering algorithm 6. Divide the data into different clusters 7. Visualise the clusters 8. Analyse the results

Technologies Used



Python



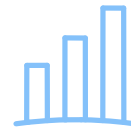
NumPy



Scikit-learn



Pandas



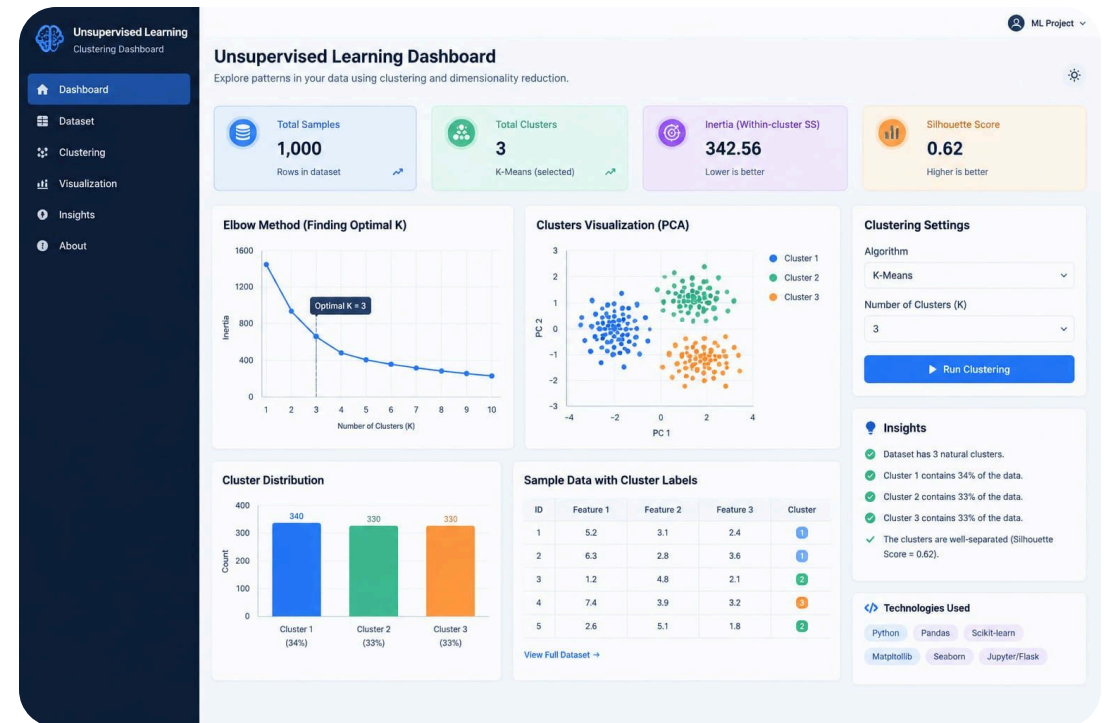
Matplotlib



Jupyter Notebook / VS Code

Project Output

The project displays different clusters created by the Machine Learning algorithm. The visualisation helps us understand how the data points are grouped based on their similarities.



Applications



Customer Segmentation



Market Analysis



Recommendation Systems



Anomaly Detection



Image Segmentation



Pattern Recognition



Data Analysis

Future Improvements

-  Using different clustering algorithms
 -  Testing different datasets
 -  Adding more visualisations
 -  Comparing K-Means with other algorithms
 -  Building a web application for the project
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Learning Outcomes

Through this project, key learnings included: basics of Unsupervised Learning, K-Means Clustering, Data Preprocessing, Data Visualisation, Python for Machine Learning, Scikit-learn, and how to analyse unlabeled data.